

Data Models, Data Preparation

*Instructor: Aditya Parameswaran**Covers Lecture(s): 9 – 10*

I Identifying Data Models

For each of the following data models, determine whether it is a matrix, relation, or dataframe.

1. What is this data model?

	gender	name	age	id
1	F	Ariel	32	12345
2	M	Devin	24	23456
3	F	Mary	28	34567
4	M	Eric	30	45678

2. What is this data model?

34	12	NaN	45
234	NaN	NaN	657
45	90	91	23
8	1	NaN	0

3. What is this data model?

City	State	Country	Zip
New York	NY	USA	10012
Berkeley	CA	USA	94702
Berkeley	CA	USA	94705

4. What is this data model?

Index	Last Name	School	Class
1	Tassens	UC Berkeley	2012
2	Lee	UCSF	2014
3	Smith	UC Berkeley	2015
4	Sanchez	UCSF	2013

II Data Model Concepts

For questions 5-8, determine whether each of the following statements is true or false. Explain.

5. All matrices are tensors and all tensors are matrices.

6. Matrices with lots of zeros are said to be sparse. Dense matrices contain very few zeros.

7. Tensors are the general term for N-dimensional matrices.

8. A tensor of shape $4 \times 5 \times 6$ has tensor rank 4.

9. Express the following relation as a matrix, and the matrix as a relation. True or False: All relations can always be expressed as matrices and vice versa. Explain

16.2	11.6	NaN
19.4	NaN	14.2

Student	Course ID	Score
Ariel	101	79.2
Ariel	102	92.1
Devin	101	85.0
Mary	103	90.0

10. Consider the following dataframe.

	Student	Course ID	Score
0	Ariel	101	79.2
1	Ariel	102	92.1
2	Devin	101	85.0
3	Mary	103	90.0

- (a) Transpose the dataframe. Assuming the original types are, in order: **String**, **Integer**, and **Float**. What are the dtypes of each transposed column?

- (b) If we transpose the dataframe again, would we get back the original dataframe? Explain.

III Pivoting and Unpivoting

For this section, we recommend you open DataHub so you can try running these queries yourself in PostgreSQL. For your convenience, we also list the questions here.

11. *Pivoting.* Write a Pandas expression to reshape the following table such that its columns are the 3 types of metals in the `CType` column, and whose values are taken from the `USD` column. Also write the resulting table. (Hint: Use `Item` as the index.)

	Item	CType	USD	EUR
0	Item0	Gold	\$1	€1
1	Item0	Bronze	\$2	€2
2	Item1	Gold	\$3	€3
3	Item1	Silver	\$4	€4

12. *Unpivoting.* Write a Pandas expression to reshape the following table from a *wide* shape into a *long* shape, using the `student` and `school` columns as identifiers, and the rest of the subjects as variables. Also write the resulting table. Hint: use `df.melt()`.

	student	school	english	math	physics
0	Andy	Z	90	60	80
1	Bernie	Y	100	5	4
2	Cindy	Z	60	99	100
3	Deb	Y	150	4	5

IV Storage Size Calculation

Consider the following schema.

```
CREATE TABLE orders(  
  student_id INTEGER, student_name VARCHAR(30), order_id INTEGER,  
  order_info VARCHAR(50), price DOUBLE,  
  FOREIGN KEY (student_id) REFERENCES students (student_id)  
);
```

13. Suppose there are 300 orders in the **orders** table. Compute the total storage cost required to store all the columns in bytes. For reference:

- CHARs are 1 byte per character.
- INTEGERS are 4 bytes.
- DOUBLES are 8 bytes.

V Minimum Description Length

Consider the relation below for this section.

Name	Species	Friendly?	NumItems
<i>TomNook</i>	<i>Raccoon</i>	1	5.0
<i>Timmy</i>	<i>Raccoon</i>	1	10
<i>Tommy</i>	<i>Raccoon</i>	1	20.0
<i>Blathers</i>	<i>Owl</i>	0	110.0
<i>Wisp</i>	<i>Ghost</i>	0	<i>Null</i>
<i>Pete</i>	<i>Pelican</i>	1	67.0
<i>Cyrano</i>	<i>Anteater</i>	0	23
<i>Mabel</i>	<i>Hedgehog</i>	1	14

14. Identify the data type for each column.

15. We will use MDL to determine the optimal type for the `NumItems` column. Assume floats are 32 bits, integers are 16 bits (although integers can also be 32 bits, let's assume we are using `smallint` here), booleans are 1 bit, and a character is 8 bits.

(a) We first consider integer type. What would be the result of the sum in the MDL equation?

(b) Now we consider float. What would be the result of the sum in the MDL equation?

(c) Now consider the string type. What would be the result of the sum in the MDL equation?

(d) Which type would MDL choose based on the calculations in the previous parts?

16. Suppose Tom Nook's `NumItems` is the string `five` instead of the number 5.0.

(a) How would this affect the MDL sum for integer type?

(b) If we do use the integer column type, how would you coerce the value `five` into an integer?

VI Data Preparation [Optional]

Consider the following table:

Year	College	Student Name
2019	L&S	Lakshya
2019	CNR	Shadaj
2020	Haas	Cathy
2020	COE	Evelyn
2020	COC	Luke
2021	CNR	Shreya
2022	L&S	Audrey
2022	COE	Alan

17. Describe how we transform a relation into a matrix and vice versa using Pandas. What functions do we use?

18. Transform the above relation to a matrix using college as columns and year as rows.

19. Is the matrix sparse or dense?

20. Transform this matrix to a sparse representation, expressed as a relation: $\begin{bmatrix} 1 & 2 & 3 & NaN \\ NaN & NaN & 7 & 8 \end{bmatrix}$